

A MATROID PROOF OF BIPARTITE CONTRACTIBILITY

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ABSTRACT. We give an independent proof that every scheme of a finite bipartite graph contains a fully rooted minor model of that graph. This proves the intended conclusion of an earlier bipartite-flow assertion of Biswal, Lee and Rao, whose prefix construction fails. The proof uses matroid union to allocate disjoint forests: either spanning trees give the model, or simultaneous component contractions yield a smaller scheme of the same target while preserving every root. When all scheme paths have at most three edges, the same construction gives branch sets of intrinsic radius at most two from their prescribed roots. A six-vertex example shows that radius one does not always suffice.

1. STATEMENT AND COLOUR NORMALISATION

All graphs are finite. Targets and hosts are simple; auxiliary multigraphs may have parallel edges. Following Kündgen, Pelsmajer and Ramamurthi [6, Definitions 1.1 and 2.1], an H -scheme in a graph G consists of distinct roots $(r_v : v \in V(H))$ and a simple r_u, r_v -path P_{uv} for every $uv \in E(H)$, with no other root internal to a path, such that any nonempty collection of paths with a common vertex has a common target endpoint. A *rooted H -minor model* is a family of pairwise disjoint nonempty connected vertex sets $(C_v : v \in V(H))$, with $r_v \in C_v$, such that C_u, C_v are adjacent whenever $uv \in E(H)$. A graph H is *contractible* if every H -scheme contains such a model.

Theorem 1.1. *Every finite simple bipartite graph is contractible.*

This answers the bipartite contractibility questions in [6, Section 8, Questions 2–4], including $K_{2,4}$, $K_{3,3}$ and bipartite theta graphs. The earlier flow assertion is discussed in Section 6. The proof uses Edmonds’ matroid union rank formula and the following part of the coloured-scheme reduction in [6, Lemma 3.3]. A minor is *root-preserving* when its vertices have pairwise disjoint connected preimages and each new root’s preimage contains its corresponding original root.

Lemma 1.2. *Every H -scheme has a root-preserving minor whose underlying graph is properly coloured by a map f to $V(H)$, with $f(r_v) = v$, and whose scheme paths use only their endpoint colours.*

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Proof. Delete unused edges and nonroot vertices. Give each root its target label. Give each nonroot an endpoint label common to all target edges whose paths contain it. Thus each path uses only its endpoint labels. Contract every connected component induced by one label. Such a component contains at most one root; the quotient colouring is proper. Each path maps to an endpoint-coloured walk. Deleting its closed excursions gives a simple path between the new roots. Another root cannot occur, since it has a different label. All paths meeting at a vertex have its label as a common target endpoint. Delete unused vertices and edges, retaining isolated prescribed roots. $\square$

Contractibility is equivalent to property (*) of Kriesell and Mohr [5, Definition 1]. The roots then form a transversal, with one vertex from each class of a proper colouring, and each required pair is joined by a path using its two colours. Such paths form a scheme; conversely, Lemma 1.2 reduces every scheme to this setting and the resulting rooted model lifts. Thus every bipartite graph has property (*).

2. A MATROID UNION EQUALITY

Let $(M_j : j \in J)$ be a finite family of multigraphs, each with distinct edge labels in a subset E_j of a finite set E . Extend its graphic matroid to E by declaring labels outside E_j to be matroid loops. Write r_j for its rank function and $M_j(X)$ for its spanning subgraph with labels in X . If $c_j(X)$ counts its components, including isolated vertices, then $r_j(X) = |V(M_j)| - c_j(X)$.

Lemma 2.1. *Let R be the maximum of $\sum_j |I_j|$ over pairwise disjoint sets $I_j \subseteq E_j$ that are forests in their respective graphs. There is $X \subseteq E$ such that*

$$(1) \quad R = |E \setminus X| + \sum_j r_j(X).$$

For any maximising family (I_j) and any such X , the forest $I_j \cap X$ spans every component of $M_j(X)$.

Proof. Edmonds' matroid union rank formula [2, Theorem 1 and the following discussion, pp. 202–203] gives

$$R = \min_{X \subseteq E} \left(|E \setminus X| + \sum_j r_j(X) \right).$$

For a minimising X and a maximising disjoint family,

$$R = \sum_j |I_j \setminus X| + \sum_j |I_j \cap X| \leq |E \setminus X| + \sum_j r_j(X) = R.$$

Equality forces $|I_j \cap X| = r_j(X)$ for every j . An independent set of this rank restricts to a spanning tree in every nontrivial component of $M_j(X)$, with isolated vertices retained. $\square$

3. A SIMULTANEOUS COMPONENT CONTRACTION

Let H be bipartite with parts (A, B) , and let G be the union of a properly endpoint-coloured H -scheme and its isolated roots, as in Lemma 1.2. Put

$$W_a = f^{-1}(a) \quad (a \in A), \quad E = f^{-1}(B) \setminus \{r_b : b \in B\}.$$

For $a \in A$, construct a multigraph M_a on W_a . Each actual occurrence of $x \in E$ on P_{ab} gives an edge labelled x between its two neighbours on that path. They are distinct vertices of colour a , so the edge is not a loop. Each label occurs at most once in M_a , since its colour determines b and the target is simple. Denote the label set of M_a by E_a .

Suppressing the internal B vertices of P_{ab} and deleting its terminal root r_b gives a path in M_a starting at r_a . These projected paths cover M_a , so M_a is connected. For an isolated target vertex, M_a is the one-vertex graph.

Lemma 3.1. *Suppose $X \subseteq E$ admits pairwise disjoint forests $F_a \subseteq E_a \cap X$, each spanning every component of $M_a(X)$. Then G has a root-preserving minor Q containing an H -scheme. If $X \neq \emptyset$, then $|V(Q)| < |V(G)|$.*

Proof. For every component K of $M_a(X)$, set

$$D_{a,K} = V(K) \cup \{x \in F_a : \text{the edge labelled } x \text{ belongs to } K\}.$$

Each allocated tree lifts to two-edge paths through its actual label vertices, making $D_{a,K}$ connected. Isolated components give singleton sets. These sets are pairwise disjoint: the original A vertices belong to distinct colour components, and every allocated B label has just one owner. They contain no B root and at most one A root.

Contract each $D_{a,K}$ to $d_{a,K}$, and delete every unallocated vertex of X . Keep the roots r_b and the vertices of $E \setminus X$ as separate vertices. Retain precisely the edges from $d_{a,K}$ to surviving B vertices y witnessed by an original edge from $V(K)$ to y , deleting other edges and parallel copies. This defines a minor Q with a proper colouring: $d_{a,K}$ receives colour a , and the surviving B vertices retain their colours. The root of a is its component vertex, and every B root is unchanged.

Fix $ab \in E(H)$. In the vertex sequence of P_{ab} , replace each A vertex by its component vertex, omit every $x \in X$, and suppress consecutive repetitions. For each omitted segment uxv , the edge labelled x places u, v in the same component of $M_a(X)$. Their images therefore coincide, regardless of which forest owns x . Each remaining step has a witnessing original edge from an A vertex to a surviving B vertex. The sequence is therefore a walk in Q between the new roots, using only colours a, b .

Delete closed excursions to obtain a simple path. No foreign A root has colour a , and the only B root retained from P_{ab} is r_b . Every intersection of the resulting paths has its colour as a common target endpoint. Thus they form an H -scheme.

The quotient has $\sum_a c_a(X) + |B| + |E \setminus X|$ vertices, so

$$|V(G)| - |V(Q)| = |X| + \sum_a r_a(X).$$

This is positive when $X \neq \emptyset$. $\square$

4. PROOF OF THEOREM 1.1

Remove isolated target vertices and their roots, restoring them as their original singleton branch sets afterwards. The result is immediate for an edgeless target. For the remaining fixed target, use strong induction on the host order. Lemma 1.2 gives a root-preserving minor K of the original host G , with a proper endpoint-coloured scheme and $|V(K)| \leq |V(G)|$.

Orient the bipartition (A, B) so that the number N_A of nonroots with colours in A is at most the number N_B with colours in B . Form the projections in Section 3 for K . They satisfy

$$\sum_{a \in A} r_a(E) = \sum_{a \in A} (|W_a| - 1) = N_A, \quad |E| = N_B.$$

Let (I_a) be a maximising disjoint family of projection forests, and let $R = \sum_a |I_a|$.

If $R = N_A$, every I_a is a spanning tree. Set

$$C_a = W_a \cup I_a \quad (a \in A), \quad C_b = \{r_b\} \quad (b \in B).$$

The lifted trees make these sets connected, and the disjoint label sets make them disjoint. Each contains its prescribed root. The final edge of P_{ab} joins r_b to W_a , giving every required contact.

Otherwise $R < N_A \leq N_B = |E|$. A minimising set X in Lemma 2.1 is nonempty, since the expression at the empty set equals $|E| > R$. That lemma supplies the forests $F_a = I_a \cap X$ required by Lemma 3.1. We obtain a root-preserving minor Q containing an H -scheme, with

$$|V(Q)| < |V(K)| \leq |V(G)|.$$

The induction hypothesis supplies a rooted model in Q . Replace its vertices by their disjoint connected preimages, first in K and then in G . This preserves connectivity, disjointness, all roots and every required adjacency. The same lift applies to the direct spanning-tree case, completing the induction. $\square$

The bipartition may reverse during induction, so lifted branch sets may expand on both original shores.

5. A SHARP RADIUS BOUND FOR SHORT PATHS

A branch set C_v has *intrinsic radius at most r at its root* if every vertex of C_v has a path of at most r edges to r_v entirely inside C_v . Distances are measured in the original host, within the branch set, rather than in a quotient graph.

Theorem 5.1. *Let H be bipartite. Every H -scheme whose paths have at most three edges contains a rooted H -minor model in which every branch set has intrinsic radius at most two at its prescribed root. The bound of two cannot be replaced by one.*

Proof. Write G_0 for the original path union together with the isolated roots. Run the proof of Theorem 1.1, keeping its disjoint connected preimages in G_0 . We maintain the following invariant: only root images have nonsingleton preimages, each of intrinsic radius at most two at its original root; every current scheme path is either a root–root edge or the image of its original three-edge path with both internal vertices still original singleton nonroots; and the quotient is properly endpoint-coloured, with surviving nonroots retaining their initial colours.

First apply Lemma 1.2. Every vertex on a monochromatic edge has a monochromatic route of at most two edges to the root of its colour: this follows by inspecting the path of at most three edges containing that edge. Thus every nontrivial monochromatic component contains its own root and has the required intrinsic radius. All other preimages are singletons. A properly bichromatic path between its differently coloured roots has odd length. If its reduced length exceeds one, it must therefore retain all three original edges and both original singleton internal vertices. This establishes the invariant.

Orient the bipartition (A, B) as before. Every current path of length greater than one is the image of an original path

$$r_a x y r_b, \quad f(x) = b, \quad f(y) = a.$$

Consequently M_a is a star centred at the image of r_a , allowing parallel edges: this path contributes the edge from that centre to y labelled x . In particular, $r_a x$ and xy are original edges of G_0 . For any $X \subseteq E$, every nontrivial component of $M_a(X)$ contains its root, and a forest spanning it allocates one label to each of its nonroot vertices. Under Lemma 3.1, a newly absorbed label is adjacent to the original root, and a newly absorbed base vertex reaches that root through its allocated label in two original edges. The old root preimage retains its existing witnesses. Only root preimages grow, so their intrinsic radii remain at most two.

The invariant also holds for the scheme paths. For the displayed demand, if $x \in X$, its projected edge puts the root and y in one component, even if x is allocated to another root or deleted. The retained edge yr_b then gives a root–root edge. This also holds if $x \notin X$ but y is absorbed. Otherwise both internal vertices stay singleton and the path is unchanged. Existing root–root edges survive the retained-edge rule. The quotient colouring is unchanged on surviving nonroots. These arguments hold after reversing the shores as well.

The two matroid-union cases complete the construction. A deficient packing gives the nonempty set X and strict decrease in host order from Lemma 3.1. A full packing absorbs every projection star into its root with

the same two-edge witnesses and supplies all required contacts. Thus the process terminates with a rooted model. The composed preimages satisfy the invariant in G_0 , so the lifted branch sets have intrinsic radius at most two. Isolated roots stay singleton.

For sharpness, take the target with edges ab, ac and a host with six distinct vertices a, b, c, a', b', c' and exactly the edges of

$$P_{ab} = a b' a' b, \quad P_{ac} = a c' a' c.$$

The paths form a scheme with prescribed roots a, b, c . Each of b, c has the unique neighbour a' . In a radius-one rooted model their disjoint branch sets cannot both contain a' , so one, say C_b , is singleton. Its contact with C_a forces $a' \in C_a$, but aa' is absent, contradicting radius one at a . Radius two is attained by $\{a, b', a'\}, \{b\}, \{c\}$. $\square$

6. THE BIPARTITE-FLOW ASSERTION

Corollary 6.1. *Let H be bipartite of minimum degree at least two. Assign its vertices to distinct terminals of a finite graph G and choose a simple path between the corresponding terminals for each target edge. If paths for edges with four distinct endpoints are vertex-disjoint, then G has an H -minor rooted at every terminal.*

Proof. A pairwise intersecting family of edges in a bipartite graph has a common endpoint: after two edges ab, ac , an edge meeting both but avoiding a would be bc , forming a triangle. Hence paths meeting at a vertex satisfy the common-endpoint condition. No terminal r_v can be internal to a path for a nonincident edge ab : at most one of a, b is adjacent to v , so v has a neighbour w outside $\{a, b\}$. Then P_{vw} and P_{ab} meet at r_v although their edges have four distinct endpoints. Thus the paths form a scheme, and Theorem 1.1 applies. $\square$

Biswal, Lee and Rao [1, Lemma 3.2] asserted this minor-existence conclusion; their Lemma 3.4 explicitly seeks to retain every terminal. The minimum-degree restriction does not distinguish their intended rooted assertion from Theorem 1.1. To remove it, attach a private four-cycle at every target vertex and a corresponding literal cycle at its host root. The enlarged scheme has a bipartite target of minimum degree two. All added host vertices are prescribed roots, so none can belong to an original root's bag. Restricting a rooted model to the original bags therefore proves contractibility of the original target.

The published intersection definition [1, p. 13:10] appears to reverse the intended preprint convention: independent demand paths must be vertex-disjoint. Correcting that wording does not repair the prefix construction on p. 13:11. For example, in the coloured $K_{2,2}$ -scheme

$$P_{ij} = a_i b'_j a'_i b_j \quad (i, j \in \{0, 1\}),$$

all eight displayed vertex names are distinct. Their left prefixes are $\{a_i\}$, and their proposed right sets are $\{b_j, b'_j, a'_0, a'_1\}$, which overlap. This refutes the construction's Lemma 3.6, not the minor-existence conclusion: a rooted model is $\{a_0, b'_0\}$, $\{a_1, b'_1\}$, $\{b_0, a'_1\}$, $\{b_1, a'_0\}$, in the order a_0, a_1, b_0, b_1 .

Lee [7, Lemma 3.3] later invoked the same bipartite-flow assertion. For some clique-flow applications, Kolbe and Spalding-Jamieson [3, Lemma 3.11 and Proposition 3.4] give another extraction argument. Using Korhonen and Lokshtanov [4, Lemmas 4.3–4.4], they choose a specific bounded-degree demand graph for each clique order and extract an unrooted clique minor. This does not establish contractibility for arbitrary bipartite targets. Our proof supplies no new spectral, separator or general bounded-depth estimate. Theorem 5.1 gives the specific intrinsic-radius bound for schemes of length at most three; it does not establish the general depth assertion of [1, Lemma 3.13], whose depth convention uses diameter in the ambient host [1, Section 1.2.2].

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